Fermionic Sign Problem with Physics-Informed Kernels

Here, we discuss the treatment of the fermionic sign problem with physics-informed kernels. The fermionic sign problem occurs in many interesting quantum systems as they are treated with Monte Carlo methods. These Monte Carlo methods ideally provide non-perturbative access to the physics of these strongly correlated systems. However, for many systems of interest at the required parameters, the ferminic sign problem occurs. It shows by a non-positive definite (and complex) measure in the path integral that leads to a bad volume scaling of the algorithms. With physics-informed kernels [3], we have shown that often a sign problem can be mitigated by finding (constructively) an integration contour in a complexified space, where the fluctuations caused by the sign problem are significantly reduced.

In this note, we discuss the explicit application of physics-informed kernels to solve fermionic sign problems. To this end, we will firstly consider the one-site Hubbard model, and later continue with further interesting systems.

1 Systems

As we defer the general discussion of physics-informed kernels to the respective papers and notes, here we will list the discussed systems, starting with the one-site Hubbard model and possibly continuing with more complex systems.

1.1 The One-Site Hubbard Model

Structure of the Theory

The one-site Hubbard model and its connection to the sign problem as it is treated by Lefschetz-thimbles has been discussed in [5]. The Hubbard model for one site has the form of

$$H = U\hat{n}_\uparrow \hat{n}_\downarrow - \mu\hat{n} \tag{1}$$

where $\hat{n}_\sigma$ denotes the number operator for spin σ and $\hat{n}$ the total density operator. The partition sum is then given by

$$\mathcal{N} = \mathrm{Tr}\left[e^{-\beta H}\right] = 1 + 2e^{\beta\mu} + e^{\beta(2\mu - U)}. \tag{2}$$

The formulation in terms of operators is not suitable for Monte Carlo methods that scale gracefully with the dimension. To that end, we consider a Hubbard Stratonovich transformation that allows us to write the partition sum as a path integral over an auxiliary field ϕ.

The action of this path integral is given by

$$S(\phi) = \frac{\beta}{2U}\phi^2 - 2\log\left[1 + e^{\beta(\mu + i\phi + \frac{1}{2}U)}\right]. \tag{3}$$

Here, the logarithm term comes from the fact that the distribution $p(\phi) = e^{-S(\phi)}$ really is a mixture of distributions as is evident from the partiton sum written in the ϕ degrees of freedom:

$$\begin{aligned}\mathcal{N} &= \sqrt{\frac{\beta}{2\pi U}}\int_{-\infty}^{\infty} d\phi\, e^{-\frac{\beta}{2U}\phi^2}\left[1 + e^{\beta(\mu + \frac{1}{2}U + i\phi)}\right]^2 \\ &= \sqrt{\frac{\beta}{2\pi U}}\int_{-\infty}^{\infty} d\phi\, e^{-\frac{\beta}{2U}\phi^2} + 2e^{\beta(\mu + \frac{1}{2}U + i\phi)}e^{-\frac{\beta}{2U}\phi^2} + e^{2\beta(\mu + \frac{1}{2}U + i\phi)}e^{-\frac{\beta}{2U}\phi^2}.\end{aligned} \tag{4}$$

From here, one can already guess that part of the interesting physics comes from the system being a mixture of distributions rather than a single distribution. With respect to the complex PIKs, we can similarly note, that the logrithm from (3) will have a non-trivial behavior in the complexified space. In terms of symmetries, the action is invariant under [5]

$$\overline{S(\phi)} = S(-\overline{\phi}) \qquad \phi \in \mathbb{C}. \tag{5}$$

The Sign Problem

In order to assess the severity of the sign problem, one can compare the true partition sum with the phase-quenched partition sum, where one considers the absolute value of the integrand, $I(\phi)$. To that end, we consider the fraction of the partition sum Z with a phase-quenched partition sum Z_{pq}. Note that while Z is invariant under legal substitutions of ϕ, Z_{pq} is not. This can simply be seen by the fact that phase differs along different contours in the complexified space.

$$Q_{sp} = \frac{\int_{-\infty}^{+\infty} d\phi I(\phi)}{\int_{-\infty}^{+\infty} d\phi |\, I(\phi)\,|}. \tag{6}$$

In [5], three different substitutions (contours) are considered that allow for an improvement of the sign problem. First, one considers the original quenched integrand. Secondly, one considers a shift of $\phi = -\varphi + 2iU$, which corresponds to a particle-hole transformation. Lastly, one considers the shift $\phi = \varphi + iU$, which corresponds to a good shift at half-filling ($\mu = \frac{1}{2}U$). At half-filling, the integrand is shifted to be completely real. In principle, one can solve all integrals directly analytically by using the Gaussian integrals

$$\int_{-\infty}^{+\infty} dx\, e^{-ax^2 + bx + c} = \sqrt{\frac{\pi}{a}}e^{\frac{b^2}{4a} + c}. \tag{7}$$

Here, for convenience, we direct compute the different phase-quenched partition sums via

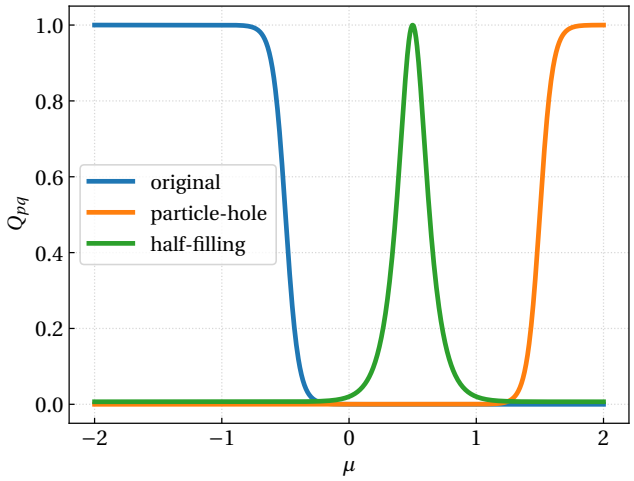


Figure 1: The fraction of the original and phase-quenched partition sum for different shifts of the integration contour. The unshifted (original) contour is shown in blue. The shift $\phi = -\varphi + 2iU$ that accords to a particle-hole symmetry is shown in orange. The shift $\phi = \varphi + iU$ that corresponds to a good shift at half-filling is shown in green. The parameters are set to $U = 1$ and $\beta = 10$ (low temperature).

numerical integration. Thies yields the situation shown in Figure 1. A value for $Q_{pq} = 1$ shows that the theory can be efficiently computed using the phase-quenched approach. However, when the value drops to 0, the theory becomes effectively intractable with direct Monte Carlo methods.

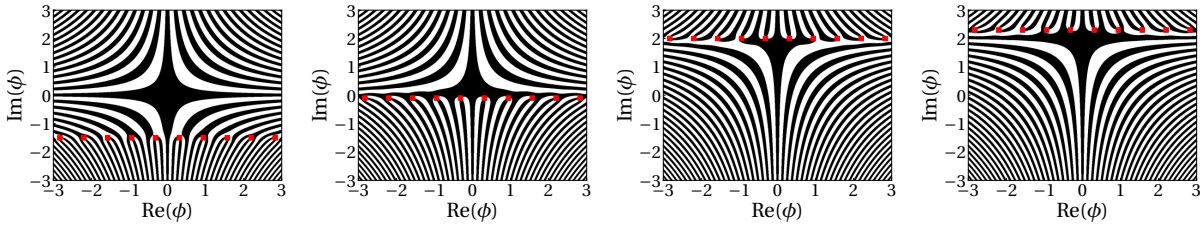
However, all these approaches somehow require some prior knowledge an engineering that is not available, especially in the case of a complete quantum field theory. The approach with the PIKs now is to find methods that generalise this notion of the contour deformation in a way that allows to find good contours somewhat automatically.

Poles

Generally, sampling from a mixture of distributions as the theory is depicted in (5) is not too difficult. One simply chooses the right Gaussians with the correct probability and then samples from the corresponding Gaussian. However, for the Hubbard model with more sites, the number of contributing Gaussian grows exponentially in some combinatorial fashion as the determinant enters the game. So, again we would like to understand the structure better on the level of the naïve action as shown in (3), which contribution the logarithm term (that will later host a determinant for more sites). However, in the complexified space, this logarithm term has poles. They can be easily identified by noting that they occur when

$$1 + e^{\beta(\mu + i\phi_\star + \frac{1}{2}U)} = 0, \tag{8}$$

as then the argument of the logarithm in the action becomes zero. Here $\phi_\star$ denotes the position of the pole. From the above equation, one can easily derive the general formula for



(a) $\mu = -2.0$. (b) $\mu = -0.6$. (c) $\mu = 1.5$. (d) $\mu = 1.8$.

Figure 2: Marking of regions where the distribution $p(\phi) \propto e^{-\hat{S}(\phi)}$ is positive in the complexified space for different values of the chemical potential μ. The parameters are set to $U = 1$ and $\beta = 10$ (low temperature).

the poles, namely

$$\mathrm{Re}(\phi_\star) = \frac{(\pi + 2\pi k)}{\beta} \quad k \in \mathbb{Z} \qquad \text{and} \quad \mathrm{Im}(\phi_\star) = \mu + \frac{1}{2}U\,. \tag{9}$$

Importantly, the position of $\mathrm{Im}(\phi_\star)$ depends on the chemical potential μ and the poles occur in a periodic fashion along the real axis.

As we will see in Figure 2, the investigation of the poles (and their evolution with μ) is indeed instructive, as they allow us to understand better the areas in which the sign problem occurs and how it occurs. In all this, we have to see that these issues occur because we treat the mixture of distributions as a single distribution. This of course is a legal move, but it may show through difficulties in the numerical approaches. Importantly, treating the mixture as one distribution allows to not deal with the humongous number of contributing Gaussians and furthermore evades the discussion of relative factors between them.

In Figure 2, we display the poles and mark the areas in black where the distribution $p(\phi)$ is positive. At $U = 1$, for $\mu < -0.5$, the poles are located in the lower half-plane and wander up as one increases μ. The point of crossing is the half-filling point. As we can see from the black lines, the distribution is positive along the real line ($\mathrm{Im}(\phi) = 0$) for $\mu < -0.5$. However, as the poles cross the real line, there is no longer a connected positive region along anywhere. This can for instance be seen for $\mu = 1.5$. Only after the poles have crossed the line for $\mathrm{Im}(\phi) = 2(U)$, there again emerges a connected line along which the distribution is positive. These observations also accord with the behavior of the quenced partition sum as shown in Figure 1.

2 Approaches

Here, now we store (largely independent) approaches to the fermionic sign problem with different methods for different systems.

2.1 Complex Physics-Informed Kernels for the One-Site Hubbard Model

As we discussed around Figure 2, the problem posed by the one-site Hubbard model is that there are regions in the complexified space where there is no connected contour (for certain parameters μ) that could be continuously deformed to the real number line again (which is required for the application of Cauchy's theorem). To begin the discussion of the approaches, we note that one can compute the solution to the PIKs analytically in this case. We start with the discussion and then move on.

2.1.1 Analytical Solution

As it turns out, to perform the analytical computation a reformulation of the Wegner equation for the physics-informed kernel $\dot{\phi}_t$ in terms of cummulative distribution functions (CDFs) $F_t(\phi)$ is instructive. This is the case, because we can represent the probability distribution function (PDF) $p_t(\phi)$ as a Gaußian mixture, for which the CDF is analytically known.

To start with this reformulation, we note that (in this zero-dimensional case), the boundary condition for the Wegner equation

$$\frac{d\,p_t(\phi)}{d\,t} = -\frac{\partial}{\partial\phi}\left(\dot{\phi}_t(\phi)\,p_t(\phi)\right) \tag{10}$$

is given by

$$\lim_{\phi\to\pm\infty} p_t\,\dot{\phi}_t(\phi) = 0\,. \tag{11}$$

Then by direction integration from $-\infty$ to ϕ by using the boundary condition and pulling the RG-time derivative out of the integral, we find

$$\frac{d}{d\,t}\int_{-\infty}^{\phi} p_t(\psi)\,d\psi = -\int_{-\infty}^{\phi}\frac{\partial}{\partial\phi}\left(\dot{\phi}_t\,p_t(\psi)\right)d\psi = -\dot{\phi}_t\,p_t(\phi) + 0\,. \tag{12}$$

Now, using that the CDF is directly given by the integral of the PDF, we can write

$$F_t(\phi) = \int_{-\infty}^{\phi} p_t(\psi)\,d\psi\,, \tag{13}$$

and thus obtain the closed formula for the physics-informed kernel $\dot{\phi}_t$ in terms of the PDF

and CDF, namely

$$\boxed{\dot{\phi}_t(\phi) = -\frac{1}{p_t(\phi)}\frac{d}{dt}F_t(\phi)\,.} \tag{14}$$

We then can further use the fact that the CDF of a PDF that is a mixture is given by the corresponding mixture of the CDFS, so

$$p_t(\phi) = \sum_i w_i\, p_{t,i}(\phi) \qquad \Rightarrow \qquad F_t(\phi) = \sum_i w_i\, F_{t,i}(\phi)\,, \qquad \text{with} \qquad \sum_i w_i = 1\,, \tag{15}$$

where $p_{t,i}(\phi)$ and $F_{t,i}(\phi)$ are the PDF and CDF of the i-th component of the mixture, respectively. In our case, the components of the mixture are Gaußians. As we know the PDF and CDF of a Gaußian analytically, one can in principle compute the physics-informed kernel $\dot{\phi}_t$ analytically via (14). Essentially, the analytical solution is given as a sum and fraction of Gaussians and error functions as error functions are the CDFs of Gaußians and are direclty able to be cotinued to the complexified space.

2.1.2 One Pikfold Solution

To compute the physics-informed kernels one must either, apply a stochastic PDE solver like a Feynman-Kac solver (which is work in progress) or expand the kernel in a suitable basis. Here, we choose the second approach. While the Feynman-Kac solver has nice properties in this aspect especially when one goes towards higher dimensions. It is important to take care of the singularities in the drift term of the respective stochastic process which is beyond the current time constraints.

Parametrisation of the action

For the remainder of this discussion, we consider a parametrisation of the action in time via

$$S(\phi) = \frac{\beta}{2U}\phi^2 - 2\log\left[1 + e^{t\beta(\mu + i\phi + \frac{1}{2}U)}\right]. \tag{16}$$

Also from a perspective of the mixture of Gaussians for the model this makes sense. Moreover, as the singularities of the theory are given by the argument of the logarithm becoming zero, turning on the oscillation via t is intuitive. Furthermore, we can as above show, that now the point of the poles behave as

$$\mathrm{Re}(\phi_\star) = -\frac{(\pi + 2\pi k)}{t\beta} \quad k \in \mathbb{Z} \qquad \text{and} \qquad \mathrm{Im}(\phi_\star) = \mu + \frac{1}{2}U\,. \tag{17}$$

Accordingly, the imaginary part of the poles is fixed by the bare couplings. However, the real part starts off at infinity and then moves towards the important regions of the distribution

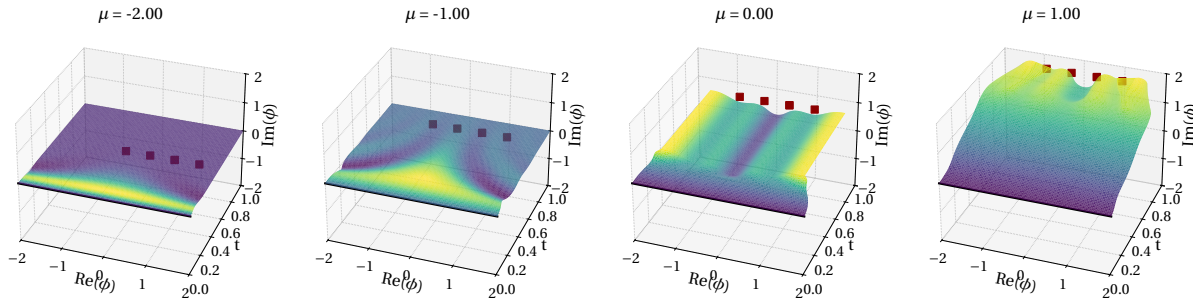


Figure 3: PIKfolds for chemical potentials μ from -2 to 1 with $U = 1$ and $\beta = 10$. The singularities of the theory at $t = 1$ are shown as red dots.

(where they are not exponentially suppressed by the enveloping real Gaußian $\exp\left[-1/2\beta/U\phi^2\right]$) as t increases.

To this end, we expand the kernel via

$$\dot{\phi}_t(\phi) = \sum_{i=0}^{N} k_{i,t} K_{i,t}(\phi) \tag{18}$$

where $K_{i,t}(\phi)$ are holomorphic or at least meromorphic functions in the complexified space. While a concrete optimisation of the basis function was not yed conducted, we consider a joint basis function of monomials and rational functions, namely

$$K_{i,t}(\phi) = \begin{cases} \frac{\phi^i}{i!} & i < N_{\text{poly}} \\ \frac{1}{\phi - \phi_\star^{(n)}(t)} & i \geq N_{\text{poly}} \end{cases}, \tag{19}$$

where we denoted $n = \pm(i - N_{\text{poly}})$ for convenience. Here, n indexes the poles as in (17). We also considered respectively normalised Hermite polynomials (both orthogonal and orthonormal in $L(\mathbb{R})$), but they performed significantly worse than the monomial basis.

Preliminary Results

In this approach we can then compute the PIKs in the given basis by optimising the coefficients $k_{i,t}$ which is easily done in this linear setting. The respective PIKFolds are shown in Figures 3 and 5. Here, we see how the PIKfolds evolve in the RG-time t for different values of the chemical potential μ. As one may expect from the discussion around Figure 1, for larger chemical potentials, the PIKfolds grow towards $2Ui$ in the imaginary direction.

One may note again that the measure on this PIKfold is completely real. Hence, one may consider whether reweighting using the PIKfold may significantly improve the sign problem (at least with respect to the original contour). To (partially) check this intuition, we show the expected occupancy in Figure 4 for different chemical potentials μ computed by reweighting using the PIKfold.

However, note that in principle, one PIKfold cannot completely cure the sign problem in the important μ regimes. This is evident from the concrete behavior of the distribution as

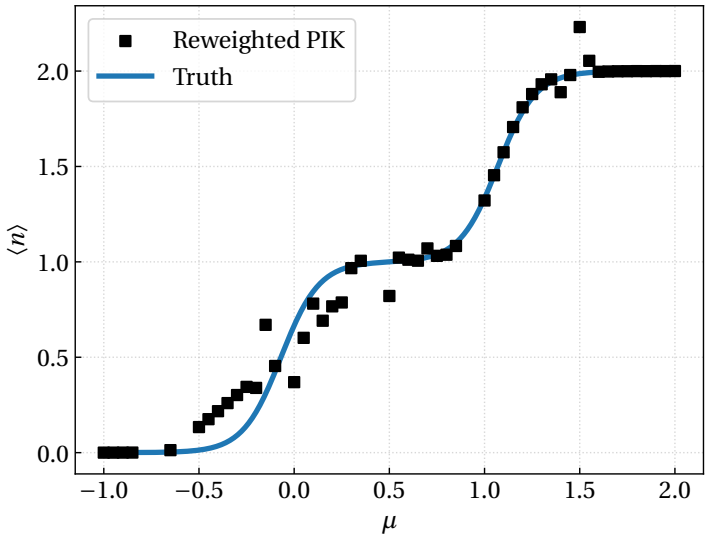


Figure 4: Expected occupancy $\langle n \rangle$ for the one-site Hubbard model with $U = 1$ and $\beta = 10$ for different chemical potentials μ. The squares show the values computed by reweighting using the PIKfold. Not RG-time extraploated and statistical errors not yet trustfully computed.

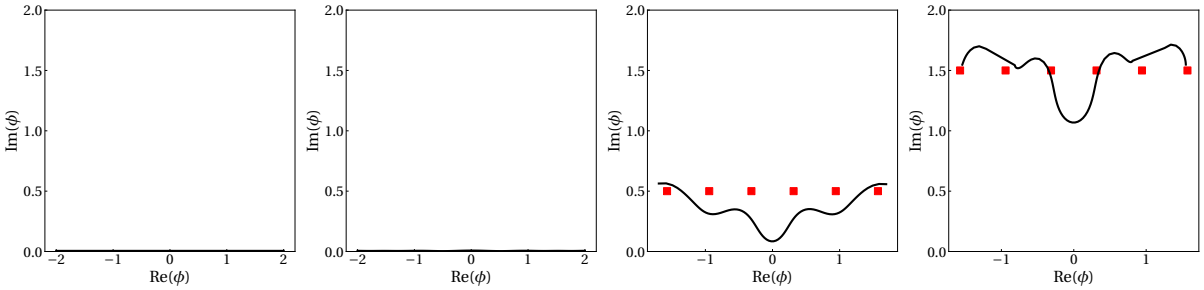


Figure 5: Final PIKfolds for chemical potentials μ from -2 to 1 with $U = 1$ and $\beta = 10$. The singularities of the theory at $t = 1$ are shown as red dots.

shown in Figure 2. This is a practical computation. However, it is also generally known (as for instance shown for the Thirring model in [4]) that contour deformations are not always able to solve a sign problem. Therefore, this one PIKfold approach can only straightforwardly soften but not cure the sign problem.

2.1.3 PIKfold Mixture

Thus, we are left with the finding that the fermionic sign problem cannot be cured by evolving real samples on a PIKfold. However, especially for the Hubbard model, we may stop and consider the possibility to write the distribution as a mixture of distribution, which then allows for multiple manifolds – one for each sub-distribution. As hinted before, for the Hubbard model, this is an especially fruitful approach. Considering also the Hubard model in a higher dimensional setting, after a suitable Hubbard–Stratonovich transformation, the ac-

tion is given roughly by [6].

$$S(\phi) = \frac{1}{2}\phi^T D\,\phi - \log\left[\det K(\phi)\det K(-\phi)\right], \tag{20}$$

where D is a given diagonal matrix and $M(\phi)$ is a given matrix that correlates the auxiliary field ϕ via

$$K_{xy}(\phi) = \delta_{xy} - e^H_{xy} e^{i\phi_x}. \tag{21}$$

This is only a rough description and the exact version can be found in [6]. Here, H denotes the hopping matrix. As can be seen, also in higher dimensions, the distribution is a mixture of (a very large number) Gaussian distributions. The problem is that the number of contributing distributions grows exponentially in some combinatorial fashion as the determinant enters the game. It nevertheless strengthens the perspective that a mixture perspective on the problem is fruitful.

To that end, we may quite generally write the distribution of the Hubbard model as as mixture model with

$$p(\phi) = \sum_k w_k \hat{\rho}_k(\phi) \qquad \text{with} \qquad \hat{\rho}_k(\phi) = e^{-\frac{1}{2}(\phi-\psi_k)^T \mathcal{C}_k^{-1}(\phi-\psi_k)}, \tag{22}$$

with the covariance matrix $\mathcal{C}_k$, the mean ψ_k, and the weights w_k. We note already here that when the unnormalised contribution of each Gaussian is given by e^{ω_k}, the the full weights are given by

$$w_k = \frac{1}{\sqrt{(2\pi)^{|\mathcal{D}|}\det \mathcal{C}_k}} \frac{e^{\omega_k}}{\sum_j e^{\omega_j}}, \tag{23}$$

where $|\mathcal{D}|$ denotes the number of lattice sites on the lattice $\mathcal{D}$. Interstingly for a fermionic theory, the fermionic nature is represented in the multitude of determinants to be computed. However, they are not dependent on field variables and hence must only be computed once and then appear in the weights w_k.

Now there are multiple methods to use this decomposition. Indeed similar approaches have been used in [1]. But see also [2, 7–9] that do similar approaches with Lefschetz thimbles. In either case, one requires a numerically tractable method to compute the respective parameters ψ_k, G_k, w_k for the contributing Gaussians. As for reasons that will become apparent only later, we limit ourselves to situations where we fit the Gaussian mixture model to capture the most important modes of the Hubbard model when the latter is not inflicted by the sign problem.

Firstly, note that as described in [3], we can map each complex Gaussian distribution $\hat{\rho}_k$

(meaning that $C_{k,xy}, \psi_{k,x} \in \mathbb{C}$) to a real Gaussian distribution $\hat{\varrho}_k$ (meaning that it has real covariance and mean $G_{k,xy}, m_{k,x} \in \mathbb{R}$) that allows for efficient sampling. Also note that the weights w_k can be chosen to be real throughout. This can always be achieved by shifts of the mean and covariance, a proof, however, is skipped here.

Now consider the Kullback–Leibler divergence between the modelled and the true distribution $\underline{p}$ and p respectively, which is given by

$$\begin{aligned}
"D_{KL}(\underline{p}\|p)" &= \int D\phi\, \underline{p}(\phi) \log \frac{\underline{p}(\phi)}{p(\phi)} \\
&= \int D\phi \sum_k w_k \underline{\hat{\varrho}}_k(\phi) \log \frac{\underline{p}(\phi)}{p(\phi)} \\
&= \sum_k \sqrt{(2\pi)^{|\mathcal{D}|} \det C_k}\, w_k \int D\phi\, \underline{\varrho}_k(\phi) \log \frac{\underline{p}(\phi)}{p(\phi)} \\
&= \sum_k \sqrt{(2\pi)^{|\mathcal{D}|} \det C_k}\, w_k \int_{\mathbb{R}} D\varphi\, \underline{\varrho}_k(\varphi) \log \frac{\underline{p}(\phi_k(\varphi))}{p(\phi_k(\varphi))} \\
&= \int_{\mathbb{R}} D\varphi \sum_k w_k \underline{\hat{\varrho}}_k(\varphi) \log \frac{\underline{p}(\phi_k(\varphi))}{p(\phi_k(\varphi))} \\
&= \mathop{\mathbb{E}}_{\substack{k\sim \mathrm{Cat}(w) \\ \varphi \sim \underline{\varrho}_k}} \left[\log \frac{\underline{p}(\phi_k(\varphi))}{p(\phi_k(\varphi))} \right]
\end{aligned} \tag{24}$$

Here, $\mathrm{Cat}(w)$ denotes the categorical distribution induced by the normalised weights w. Most importantly, we used the map $\phi_k(\varphi)$ which represents the mapping between the complex Gaussian ρ_k to the real Gaussian ϱ_k. This allows to optimise parameters of the modelled distribution $\underline{p}$ to best match the true distribution p which is itself a mixture of Gaussians. At best one reads (25) from the last line to the top. This gives a precise meaning to the measure $D\phi\underline{p}(\phi)$ in the first line in terms of the mixtures. Also note for the concrete example of the Hubbard model that we can further write the KL divergence as

$$"D_{KL}(\underline{p}\|p)" = \mathop{\mathbb{E}}_{\substack{k\sim \mathrm{Cat}(w) \\ \varphi \sim \underline{\varrho}_k}} \left[\frac{1}{2}\phi_k^T(\varphi) D\, \phi_k(\varphi) - \frac{1}{2}\phi_k^T(\varphi) \underline{D}\, \phi_k(\varphi) - \log \frac{\det M(\phi_k(\varphi))}{\det \underline{M}(\phi_k(\varphi))} \right]. \tag{25}$$

Here, we could directly assume form the known action that the enveloping Gaussian is given by the specific Gaussian part $\phi_k^T D \phi_k$. Here, the $\det M$ from the explicit theory is truly given by a determinant. However, the $\det \underline{M}$ here can be cheaply computed from the model parameters as one has broken the system in a finite mixture of Gaussians. In our explicit example of the one-site Hubbard model, the concrete formula is given by

$$"D_{KL}(\underline{p}\|p)" = \mathop{\mathbb{E}}_{\substack{k\sim \mathrm{Cat}(w) \\ \varphi \sim \underline{\varrho}_k}} \left[\frac{1}{2}\left(\frac{1}{\sigma^2} - \frac{1}{\underline{\sigma}^2} \right) \phi_k^2(\varphi) - \log \frac{\det M(\phi_k(\varphi))}{\det \underline{M}(\phi_k(\varphi))} \right] + \log \mathcal{N} - \log \underline{\mathcal{N}} \tag{26}$$

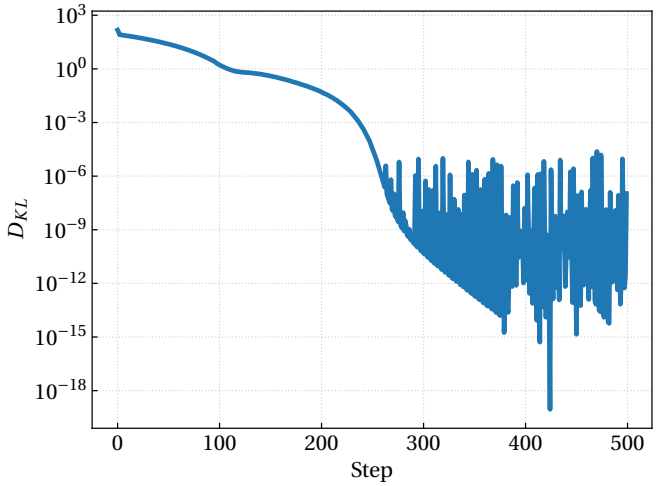


Figure 6: History of the KL divergence during the optimisation of the Gaussian mixture model for the one-site Hubbard model with $U = 1$, $\beta = 10$, and $\mu = 1$. Note that here the forward KL divergence was used. This has to be updated with the reverse.

with the respecitvely chosen variance $\sigma^2 = U/\beta$. Moreover, the determinant terms are straightforwardly given by

$$\det M(\phi) = 1 + 2e^{\beta(\mu+\frac{1}{2}U+i\phi)} + e^{2\beta(\mu+\frac{1}{2}U+i\phi)} \tag{27}$$
$$\det \underline{M}(\phi) = \sum_k \exp\left[i\phi \frac{\underline{m}^2{}_k}{\underline{\sigma}^2} + \frac{1}{2}\frac{\underline{m}^4_k}{\underline{\sigma}^2} + \underline{\omega}_k \right]$$

This form of the modelled determinant term $\det \underline{M}$ of the one-site Hubbard model relates to the modelled distribution $\underline{p}$ as the following parametrised mixture of Gaussians

$$\underline{p}(\phi) = \frac{1}{\underline{\mathcal{N}}} \sum_k e^{\underline{\omega}_k} e^{-\frac{1}{2\underline{\sigma}^2}\left(\phi - i\underline{m}_k{}^2\right)^2}, \tag{28}$$

for real parameters $\underline{\sigma}_k, \underline{m}_k, \underline{\omega}_k$. While the "KL divergence" can be mapped to a mixture distribution that has no sign problem as we showed in (25), still the observable is a complex one as the integrand is complex. Consequently, optimisation is impossible as we require a real optimisation criterion to have zeros in our optimisation landscape only when the parameters are optimal. To that end, one could consider the optimisation criterion $|D_{KL}|^2$. This can again be efficiently computed without a sign problem. However, it does not necessarily have a zero for only the optimal parameters. An instructive choice is thus

$$"D^{|\cdot|}_{KL}(\underline{p}\|p)" = \mathop{\mathbb{E}}_{\substack{k\sim \mathrm{Cat}(w) \\ \varphi\sim\underline{\varrho}_k}} \left[|\underline{S}(\phi_k(\varphi)) - S(\phi_k(\varphi))|^2 \right]. \tag{29}$$

As the complex integrand does not cancel out here, the above divergence has (experimen-

tally) shown to have the correct global zero in parameter space. An explicit example for the behavior of the criterion during optimisation is shown in Figure 6. By computing the reweighting factor and the density we can further check the quality of the fit. At this point we can already note that similar approaches can be conducted for the Fisher divergence, for which the above considerations appear more naturally.

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